

Results. From Fig. 4 (and expanded in Fig. 8), we observe that across datasets, pretraining (PT) yields competitive (often superior) OOD calibration with respect to end-to-end (E2E) G- Δ UQ. With the exception of GOODMotif (basis) dataset, PT G- Δ UQ improves the OOD ECE over both vanilla and E2E G- Δ UQ models at comparable or improved OOD accuracy (6/8 datasets). Furthermore, PT G- Δ UQ also improves the ID ECE on all but the GOODMotif(size) (6/8), where it performs comparably to the vanilla model, and maintains the ID accuracy. Notably, as only an anchored classifier is trained, PT G- Δ UQ substantially reduces training time relative to E2E G- Δ UQ and vanilla models (see App. A.13), highlighting its strengths as a light-weight, effective strategy for improving uncertainty estimation.

7 CONCLUSION

We propose G- Δ UQ, a novel training approach that adapts stochastic data centering for GNNs through newly introduced graph-specific anchoring strategies. Our extensive experiments demonstrate G- Δ UQ improves calibration and uncertainty estimates of GNNs under distribution shifts.

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REFERENCES

- Uri Alon and Eran Yahav. On the bottleneck of graph neural networks and its practical implications. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2021. [Pg. 9.]
- Rushil Anirudh and Jayaraman J. Thiagarajan. Out of distribution detection via neural network anchoring. In *Asian Conference on Machine Learning, ACML 2022, 12-14 December 2022, Hyderabad, India, 2022*. [Pg. 3.]
- Beatrice Bevilacqua, Yangze Zhou, and Bruno Ribeiro. Size-invariant graph representations for graph classification extrapolations. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2021. [Pg. 23.]
- Charles Blundell, Julien Cornebise, Koray Kavukcuoglu, and Daan Wierstra. Weight uncertainty in neural network. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2015. [Pg. 2.]
- Davide Buffelli, Pietro Liò, and Fabio Vandin. Sizeshiftreg: a regularization method for improving size-generalization in graph neural networks. In *Proc. Adv. in Neural Information Processing Systems (NeurIPS)*, 2022. [Pg. 6 and 23.]
- Gabriele Corso, Luca Cavalleri, Dominique Beaini, Pietro Liò, and Petar Velickovic. Principal neighbourhood aggregation for graph nets. In *NeurIPS*, 2020. [Pg. 6.]
- Nicki Skafte Detlefsen, Jiri Borovec, Justus Schock, Ananya Harsh, Teddy Koker, Luca Di Liello, Daniel Stancl, Changsheng Quan, Maxim Grechkin, and William Falcon. Torchmetrics - measuring reproducibility in pytorch, 2022. URL <https://github.com/Lightning-AI/torchmetrics>. [Pg. 28.]
- Mucong Ding, Kezhi Kong, Jiuhai Chen, John Kirchenbauer, Micah Goldblum, David Wipf, Furong Huang, and Tom Goldstein. A closer look at distribution shifts and out-of-distribution generalization on graphs. In *NeurIPS 2021 Workshop on Distribution Shifts: Connecting Methods and Applications*, 2021. [Pg. 9 and 19.]

- Vijay Prakash Dwivedi, Chaitanya K. Joshi, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. Benchmarking graph neural networks. *CoRR*, 2020. [Pg. 9.]
- Vijay Prakash Dwivedi, Anh Tuan Luu, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. Graph neural networks with learnable structural and positional representations. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2022a. [Pg. 4 and 9.]
- Vijay Prakash Dwivedi, Ladislav Rampásek, Michael Galkin, Ali Parviz, Guy Wolf, Anh Tuan Luu, and Dominique Beaini. Long range graph benchmark. In *Proc. Adv. in Neural Information Processing Systems NeurIPS, Datasets and Benchmark Track*, 2022b. [Pg. 9.]
- Saurabh Garg, Sivaraman Balakrishnan, Zachary C. Lipton, Behnam Neyshabur, and Hanie Sedghi. Leveraging unlabeled data to predict out-of-distribution performance. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2022. [Pg. 29.]
- Thomas Gaudelot, Ben Day, Arian R. Jamasb, Jyothish Soman, Cristian Regep, Gertrude Liu, Jeremy B. R. Hayter, Richard Vickers, Charles Roberts, Jian Tang, David Roblin, Tom L. Blundell, Michael M. Bronstein, and Jake P. Taylor-King. Utilising graph machine learning within drug discovery and development. *CoRR*, abs/2012.05716, 2020. [Pg. 1.]
- Shurui Gui, Xiner Li, Limei Wang, and Shuiwang Ji. GOOD: A graph out-of-distribution benchmark. In *Proc. Adv. in Neural Information Processing Systems (NeurIPS), Benchmark Track*, 2022. [Pg. 5, 7, 23, and 30.]
- Devin Guillory, Vaishaal Shankar, Sayna Ebrahimi, Trevor Darrell, and Ludwig Schmidt. Predicting with confidence on unseen distributions. In *ICCV*, 2021. [Pg. 29.]
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proc. of the Int. Conf. on Machine Learning, (ICML)*, 2017. [Pg. 1, 2, 9, and 28.]
- Kartik Gupta, Amir Rahimi, Thalaiyasingam Ajanthan, Thomas Mensink, Cristian Sminchisescu, and Richard Hartley. Calibration of neural networks using splines. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2021. [Pg. 1, 2, and 28.]
- Arman Hasanzadeh, Ehsan Hajiramezanali, Shahin Boluki, Mingyuan Zhou, Nick Duffield, Krishna Narayanan, and Xiaoning Qian. Bayesian graph neural networks with adaptive connection sampling. In *ICML*, 2020. [Pg. 2.]
- Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2017. [Pg. 1, 2, and 8.]
- Dan Hendrycks, Mantas Mazeika, and Thomas G. Dietterich. Deep anomaly detection with outlier exposure. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2019. [Pg. 2, 7, and 8.]
- Dan Hendrycks, Nicholas Carlini, John Schulman, and Jacob Steinhardt. Unsolved problems in ML safety. *CoRR*, abs/2109.13916, 2021. [Pg. 7.]
- Dan Hendrycks, Steven Basart, Mantas Mazeika, Andy Zou, Joseph Kwon, Mohammadreza Mostajabi, Jacob Steinhardt, and Dawn Song. Scaling out-of-distribution detection for real-world settings. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2022a. [Pg. 8.]
- Dan Hendrycks, Andy Zou, Mantas Mazeika, Leonard Tang, Bo Li, Dawn Song, and Jacob Steinhardt. Pixmix: Dreamlike pictures comprehensively improve safety measures. In *Proc. Int. Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2022b. [Pg. 1 and 7.]
- José Miguel Hernández-Lobato and Ryan P. Adams. Probabilistic backpropagation for scalable learning of bayesian neural networks. In *Proc. Int. Conf. on Machine Learning (ICML)*, 2015. [Pg. 2.]
- Hans Hao-Hsun Hsu, Yuesong Shen, Christian Tomani, and Daniel Cremers. What makes graph neural networks miscalibrated? In *Proc. Adv. in Neural Information Processing Systems NeurIPS*, 2022. [Pg. 1, 2, and 28.]

- Eyke Hüllermeier and Willem Waegeman. Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods. *Mach. Learn.*, 110(3), 2021. [Pg. 3.]
- Yiding Jiang, Dilip Krishnan, Hossein Mobahi, and Samy Bengio. Predicting the generalization gap in deep networks with margin distributions. In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6-9, 2019*. OpenReview.net, 2019. [Pg. 1, 2, 7, and 29.]
- Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *ICLR*, 2017. [Pg. 6.]
- Konstantin Kirchheim, Marco Filax, and Frank Ortmeier. Pytorch-ood: A library for out-of-distribution detection based on pytorch. In *Workshop at the Proc. Int. Conf. on Computer Vision and Pattern Recognition CVPR*, 2022. [Pg. 8.]
- Boris Knyazev, Graham W. Taylor, and Mohamed R. Amer. Understanding attention and generalization in graph neural networks. In *Proc. Adv. in Neural Information Processing Systems (NeurIPS)*, 2019. [Pg. 9.]
- Meelis Kull, Miquel Perelló-Nieto, Markus Kängsepp, Telmo de Menezes e Silva Filho, Hao Song, and Peter A. Flach. Beyond temperature scaling: Obtaining well-calibrated multi-class probabilities with dirichlet calibration. In *Proc. Adv. in Neural Information Processing Systems NeurIPS*, 2019. [Pg. 1, 2, and 28.]
- Ananya Kumar, Percy Liang, and Tengyu Ma. Verified uncertainty calibration. In *Proc. Adv. in Neural Information Processing Systems NeurIPS*, 2019. [Pg. 28.]
- Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Proc. Adv. in Neural Information Processing Systems (NeurIPS)*, 2017. [Pg. 2.]
- Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Proc. Adv. in Neural Information Processing Systems NeurIPS*, 2018. [Pg. 8.]
- Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Proc. Adv. in Neural Information Processing Systems NeurIPS*, 2020. [Pg. 8.]
- Matthias Minderer, Josip Djolonga, Rob Romijnders, Frances Hubis, Xiaohua Zhai, Neil Houlsby, Dustin Tran, and Mario Lucic. Revisiting the calibration of modern neural networks. In *Proc. Adv. in Neural Information Processing Systems (NeurIPS)*, 2021. [Pg. 9.]
- Christopher Morris, Nils M. Kriege, Franka Bause, Kristian Kersting, Petra Mutzel, and Marion Neumann. Tudataset: A collection of benchmark datasets for learning with graphs. In *ICML 2020 Workshop on Graph Representation Learning and Beyond (GRL+ 2020)*, 2020. URL www.graphlearning.io. [Pg. 6.]
- Luis Müller, Mikhail Galkin, Christopher Morris, and Ladislav Rampásek. Attending to graph transformers. *CoRR*, abs/2302.04181, 2023. [Pg. 9.]
- Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using bayesian binning. In *Proc. Conf. on Adv. of Artificial Intelligence (AAAI)*, 2015. [Pg. 28.]
- Aviv Netanyahu, Abhishek Gupta, Max Simchowitz, Kaiqing Zhang, and Pulkit Agrawal. Learning to extrapolate: A transductive approach. In *Proc. Int. Conf. on Learning Representations (ICLR)*, 2023. [Pg. 3.]
- Nathan Ng, Neha Hulkund, Kyunghyun Cho, and Marzyeh Ghassemi. Predicting out-of-domain generalization with local manifold smoothness. *CoRR*, abs/2207.02093, 2022. [Pg. 29.]
- Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model’s uncertainty? evaluating predictive uncertainty under dataset shift. In *Proc. Adv. in Neural Information Processing Systems NeurIPS*, 2019. [Pg. 1, 2, 5, 7, and 28.]